All devices at the time of its mounting associate itself with some respective file system which further attaches itself to the directory system. The three significant steps which are to be followed to create file system in Linux are as follows –

Step I: Use fdisk command or Disk Utility to create partitions in the storage volume

Step II: Use mkfs command to format the partitions created in Step I

Step III: Use the mount command to mount the partitions or it can be automated using the /etc/fstab file

In case you have added a new disk to your Linux terminal, the following steps will be guiding you to its mounting stage –

Let us assume that we have a new partition named /dev/sda1 has been created.

Step I: You can check whether the Linux kernel can identify the partition, or not by using the command cat /proc/partitions

Step II: Depending upon the type of the device or file system, decide from the options which gets displayed on typing the command

```
[root@localhost ~]# mkfs.<tab><tab>
```

In the above command, you may use filesystem types as mentioned below: mkfs.btrfs mkfs.cramfs mkfs.ext2 mkfs.ext3 mkfs.ext4 mkfs.minix mkfs.xfs

mkfs (MakeFileSystem) is a command used to create a new filesystem.

For instance, I want to have .ext4 file type. It can be done in the following way (output may differ due to device type or specifications) –

```
[root@localhost ~]# mkfs.ext4  /dev/sda1
mke2fs 1.42.9 (20-Feb-2019)
Filesystem label=
OS type: Linux
Block size=4096 (log=2)
Fragment size=4096 (log=2)
Stride=0 blocks, Stripe width=8191 blocks
194688 inodes, 778241 blocks
38912 blocks (5.00%) reserved for the superuser
First data block=0
Maximum filesystem blocks=799014912
24 block groups
32768 blocks per group, 32768 fragments per group
8112 inodes per group
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376, 294912
```

Allocating group tables: done

Writing inode tables: done